Air University
Final Examination: Fall 2024

Department of Electrical and Computer Engineering

Subject: Calculus and Analytical Geometry (MA-110) **Total Marks:** 100
Date: **Time:**
Course Instructor: Dr. Waseem Khan **Classes:** BEET-I-A, BCE-I-A
Student Roll Number:_____

Special Instructions:
1. This is a closed book/closed notes examination. It contains ***03*** questions spanned over 02 page(s). Make sure that you have got the complete question paper.
2. **CALCULATORS ARE NOT ALLOWED.**

Question # 01: [CLO 01, PLO 1] **Marks: 15+5+5+5**

Based on your **understanding** of functions and limits, answer the following questions:

a. Consider a function defined as $g(x) = 5 - 2x$ and another function $f(x)$ sketched in Fig. 1.
   i. Evaluate $f(-1)$.
   ii. Evaluate $f(g(2))$.
   iii. Find $\lim_{x\to 1^+} f(x)$ and $\lim_{x\to 1^-} f(x)$

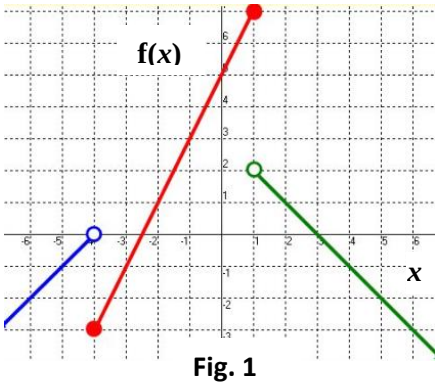


Fig. 1

b. Write down the domain and range of the function:

$$y(t) = \frac{e^{-2t}-1}{2},\ t \geq 0$$

c. Sketch the graph of the function $h(x) = \frac{1}{x-1}$

d. Evaluate the limit: $\lim_{y\to 0} \frac{5y^3 + 8y^2}{3y^4 - 16y^2}$

Question # 02: [CLO 02, PLO 2] **Marks: 10+15+15**

Applying the concepts of derivatives, answer the following questions.

a. Consider the curve given by the following relation of x and y. Find the equation of the tangent on the curve at the given point. $x^2 \cos^2 y - \sin y = 0, \quad (0, \pi)$

b. A particle is moving along x-axis and its velocity-time graph is shown in Fig. 2.
   i. Identify the time instants when the particle is not moving.
   ii. Identify the time instants when the particle is not accelerating.
   iii. Identify the time instants when the particle changes its direction.

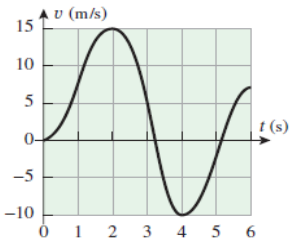


Fig. 2

c. A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec.

i. How fast is the top of the ladder sliding down the wall then?

ii. At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?

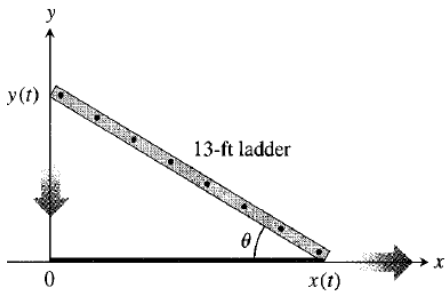


Fig. 3

Question # 03: [CLO 02, PLO 1] **Marks: 20+10**

Apply the concepts and techniques of integration to answer the following questions.

a. Evaluate the integrals given below. Use substitution where suggested.

i. $$\int \left(1-\cos\frac{t}{2}\right)^2 \sin\frac{t}{2}\,dt, \quad u = 1-\cos\frac{t}{2}$$

ii. $$\int_0^2 \left(\frac{t^2}{4}-7t+5\right) dt.$$

b. Find the area of the shaded region in Fig. 4.

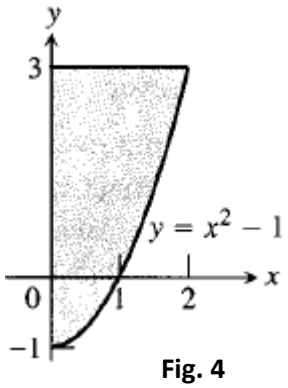


Fig. 4